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Research Update

Two-sensor post-selection for the metrology paper, the hidden manifold model revisited, and a review of four weeks of work

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2026-07-16 · Lab Meeting

Outline

1. **Post-selection: the two-sensor extension.** Where the paper stands, and the new numerical results: independent product filters stay additive, while a collective filter reaches the full $4\times$ conditional QFI, from entangled and separable initial states alike.
2. **The hidden manifold model.** The journal-club material revisited: structured data from a generator, Gaussian equivalence, closed learning dynamics, and what has moved since June 29.
3. **What I have done.** A log-derived look at where the last four weeks actually went, checked against the June plan.

1 · RESULTS 2 · MODEL & THEORY 3 · REVIEW

01

Post-selection: the two-sensor extension

Product versus collective filters, and the conditional QFI

RECAP The paper in one question

How should a post-selection filter be designed to maximize the signal-domain conditional QFI in spin-qutrit sensing under dephasing?

The single-sensor part is settled (shown in the June 18 and July 2 meetings):

- The field writes a phase $\phi = \gamma_e B t_s$ into the sensing qubit while dephasing shrinks its coherence, $\eta = e^{-t_s/T_2}$.
- After encoding, a **filter** $K = D_\gamma V$ acts: a rotation V, then a partial-strength coupling of the sensing levels to the auxiliary $|{+1}\rangle$ level; only runs that stay in the sensing subspace are kept ("conditional" = per accepted event).
- The **matched** (state-aligned) filter and its operating point are closed-form, and they recover the QFI that dephasing had erased.

The storyline: **(1)** spin-qutrit post-selection under dephasing, **(2)** phase-QFI versus signal-QFI, **(3)** state-filter alignment and the optimal filter, **(4)** the NV proof-of-principle experiment, **(5)** the readout-limited sensitivity advantage, and **(6) the collective two-sensor extension (my part, today).**

DESIGN Two sensors: product versus collective filters

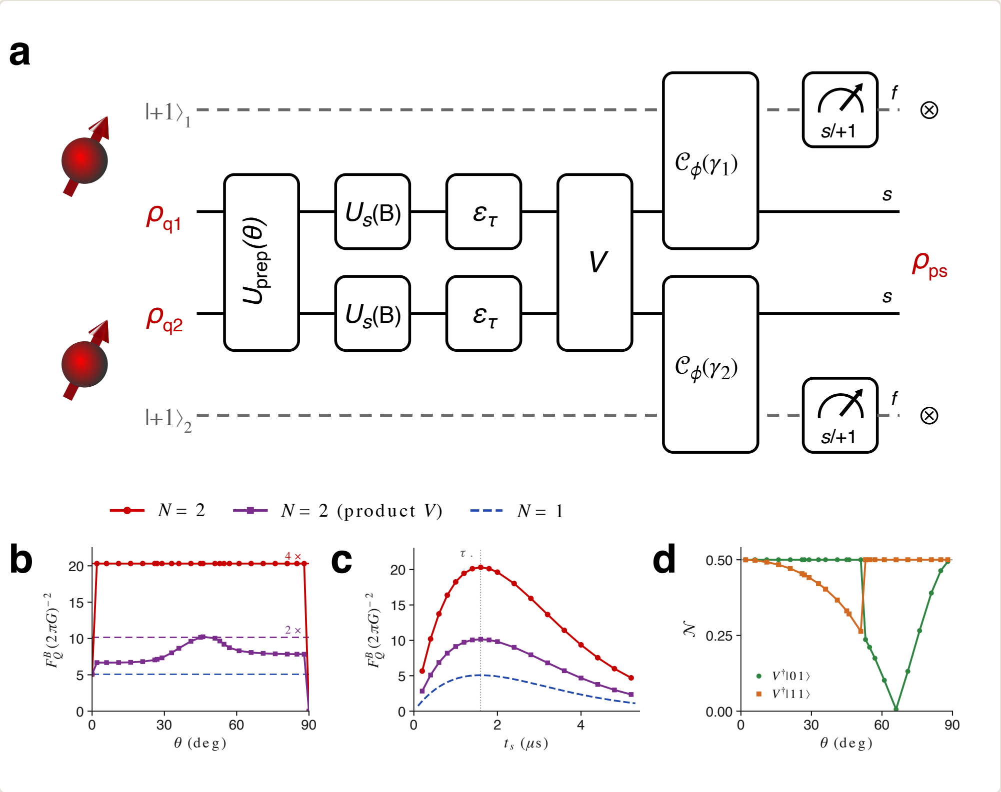
With two sensors, the single-sensor recipe has two natural extensions, and they are not equal.

- **Product filter** $K_1 \otimes K_2$: apply the single-sensor optimal filter to each sensor independently. The conjugator $V = V_1 \otimes V_2$ is local, so each sensor is asked its own question and the answers are combined classically.
- **Collective filter** $K_{\rm coll} = D_\gamma V$ with an entangling V: the success branch of the joint circuit reduces to one effective Kraus operator that does **not** factorize, $K_{\rm coll} \neq K_1 \otimes K_2$. The two sensors are asked a single joint question.

Both classes are optimized on equal footing: at every initial-state angle θ, a structure-free numerical search over the conjugator, both post-selection strengths γ_1, γ_2, and the sensing time t_s (up to 18 parameters, multi-seed annealing plus polish).

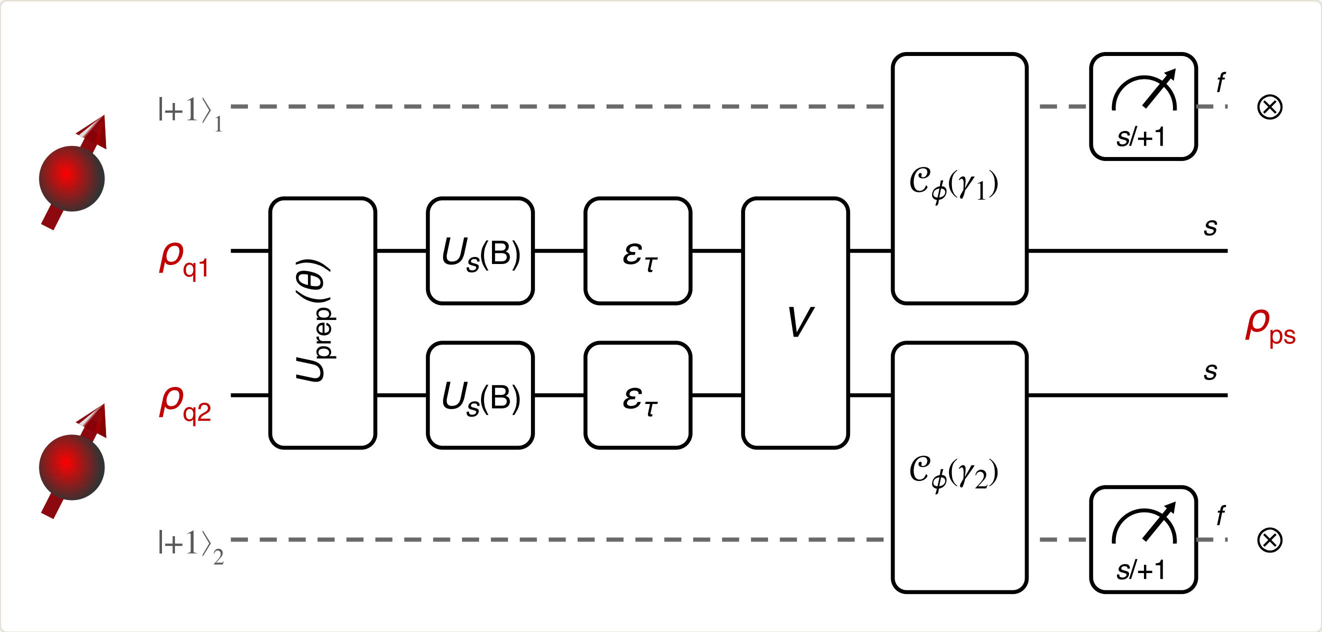
Preview of the result. The product class stays *additive*: its conditional QFI is the sum of the two single-sensor contributions. The collective class reaches about $4\times$ the single-sensor optimum, and it does so for separable initial states too.

RESULTS The figure for the paper



The two-sensor composite: ***(a)*** *the scheme,* ***(b)*** *conditional QFI versus* θ*,* ***(c)*** *conditional QFI versus sensing time,* ***(d)*** *the filter's accepted-axis entanglement. Panel walk-through on the next four slides.*

SCHEME Panel (a): the two-sensor post-selection circuit

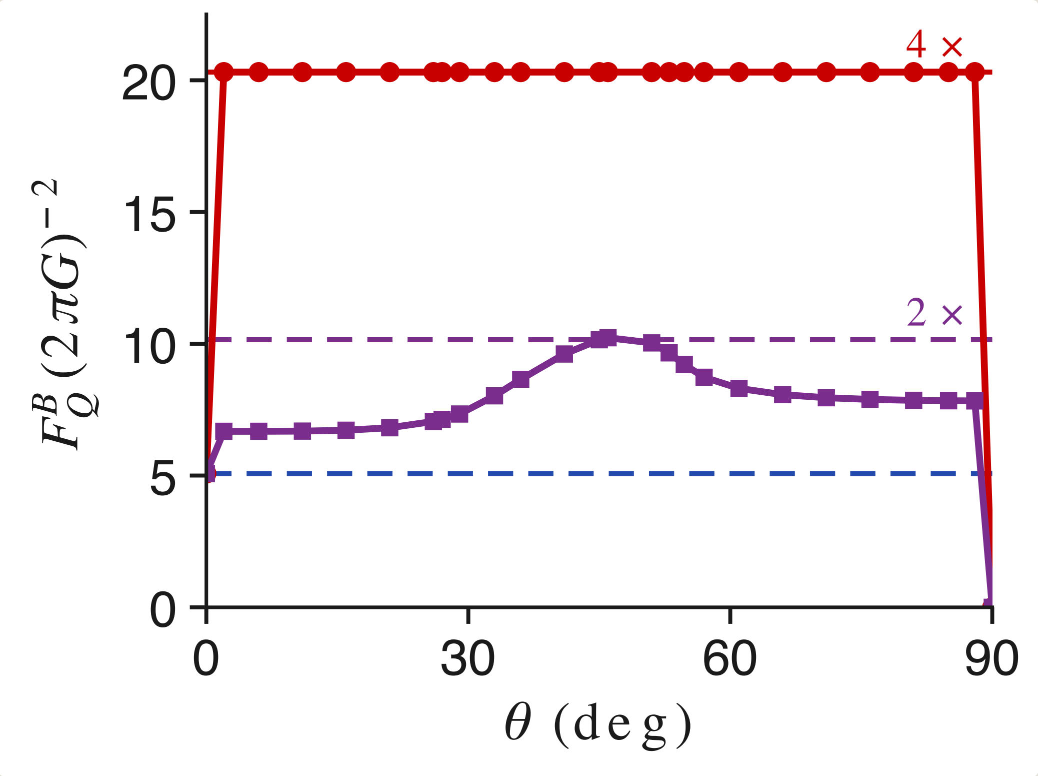


- $U_{\text{prep}}(\theta)$ prepares the two-sensor probe; $U_s(B)$ writes the field phase $\phi = \gamma_e B\, t_s$ into each sensor while the dephasing channel ε_τ shrinks the coherence to $\eta = e^{-t_s/T_2}$.
- V is the trainable **conjugator**, the only place the two sensors meet; $\mathcal{C}_\phi(\gamma_i)$ couples sensor i to its $|+1\rangle$ level with post-selection strength γ_i (weak up to projective). In the numerics each qutrit is a sensing qubit plus one meter qubit.
- Reading out $s/+1$ heralds the run: f outcomes are discarded, and the doubly successful branch is the effective Kraus operator $K = D_{\gamma_1\gamma_2}V$; we maximize the conditional QFI of ρ_{ps} per accepted event over $\{V, \gamma_1, \gamma_2, t_s\}$.

RESULTS

Panel (b): collective $4\times$, product additive

- The **collective filter** (red) reaches a flat plateau $F_Q^B = 20.31$ in the figure's units at every interior θ: exactly $4\times$ the single-sensor optimum 5.08 (blue), which is the N^2 value at $N = 2$.
- The best **product filter** (violet) is *additive*: its conditional QFI is the sum of the two single-sensor contributions. It touches its $2\times$ ceiling only at the symmetric point $\theta = 45^\circ$ and stays below elsewhere.
- The plateau is θ-independent, and in further numerics an optimized collective filter reaches the same $4\times$ from **separable** initial states: the filter itself supplies the entanglement. So initial-state entanglement is **not a prerequisite** for the maximal enhancement.

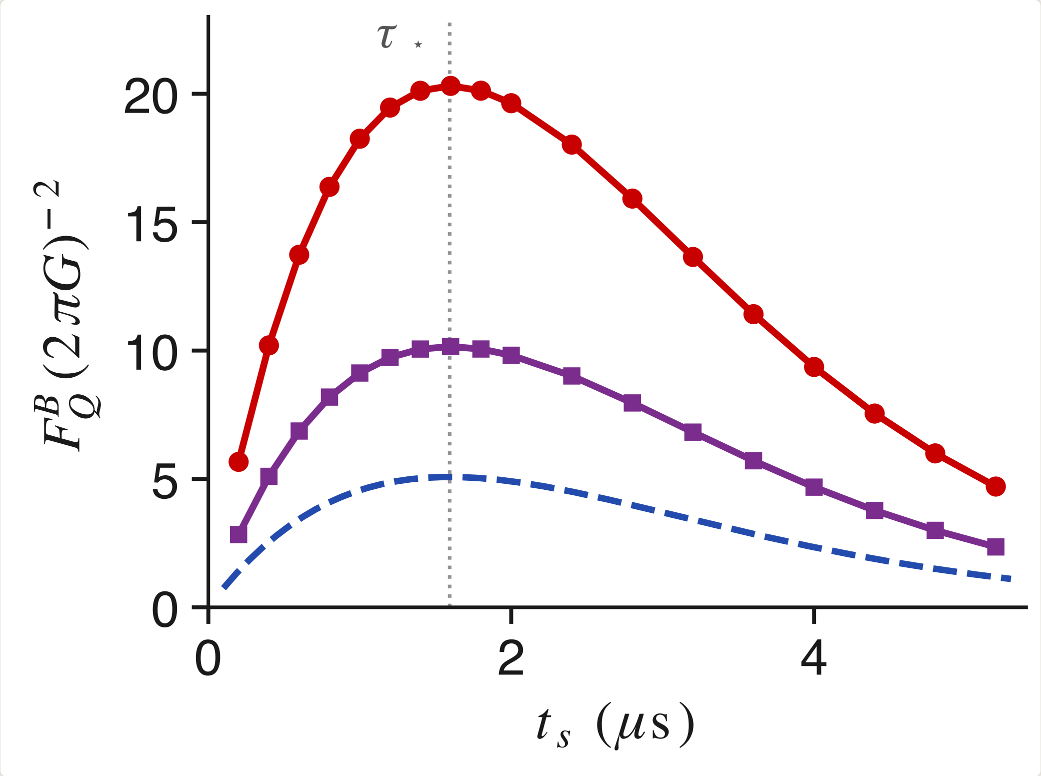


Conditional QFI versus the initial-state angle θ at the optimal sensing time. Dashed lines mark $1\times, 2\times, 4\times$ the single-sensor optimum; at the edges $\theta = 0, 90^\circ$ the probe family degenerates and the QFI collapses.

RESULTS

Panel (c): the optimal post-selection time

- Sweeping the sensing time gives an **interior optimum** $t_s = \tau_\star \approx 0.797\,T_2 \approx 1.59\ \mu\text{s}$ (here $T_2 = 2\ \mu\text{s}$): waiting longer accumulates more phase, $\phi = \gamma_e B\,t_s$, but costs coherence, $\eta = e^{-t_s/T_2}$, and $\tau_\star$ balances the two.
- The signal-domain QFI behaves as $F_Q^B \propto (\gamma_e t_s)^2\,\eta^2/(1-\eta^2)$ at the matched filter, which peaks where $1-\eta^2 = t_s/T_2$; this is why the phase-QFI versus signal-QFI distinction matters in the paper.
- All three curves peak at the **same** $\tau_\star$, and the product-filter curve is exactly half the collective one at every t_s: the $4\times$ separation is **pointwise in time**, not a fine-tuned coincidence of the operating point.

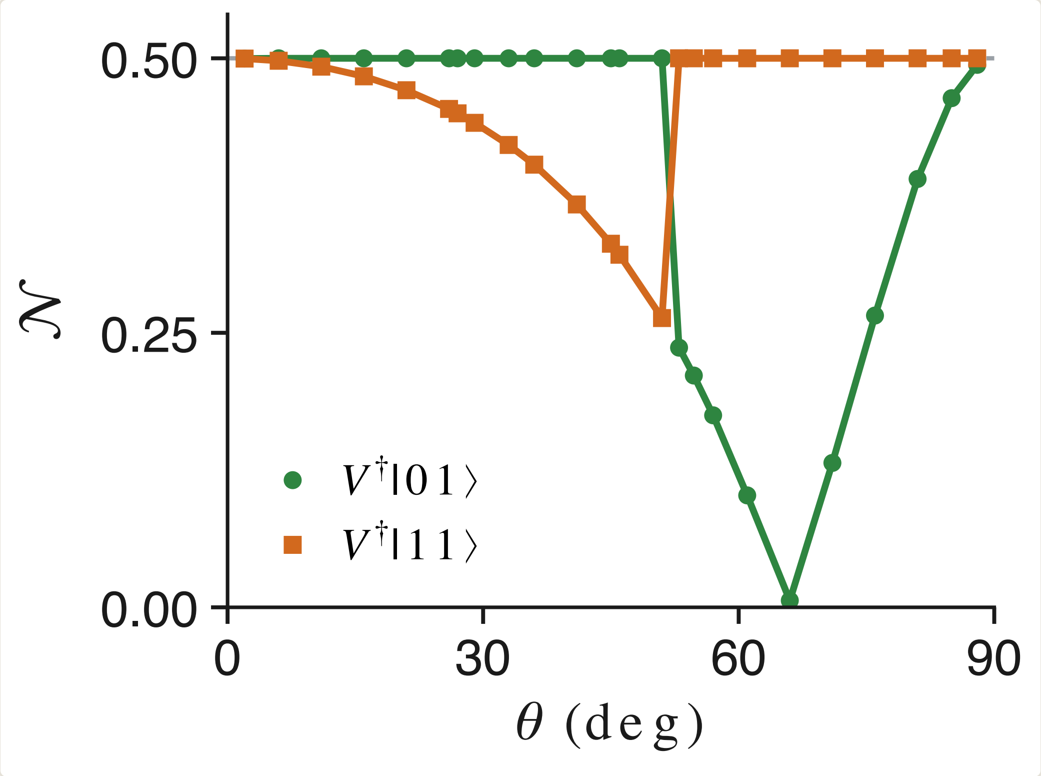


Conditional QFI versus sensing time t_s, filter re-optimized at each point; the dotted line marks $\tau_\star$.

RESULTS

Panel (d): the optimal filter is intrinsically two-qubit

- At each θ the conjugator V is taken **from the structure-free optimizer itself** (no ansatz), and we compute the entanglement of the two accepted filter axes $V^\dagger|01\rangle$ and $V^\dagger|11\rangle$.
- **At least one accepted axis is a Bell state** ($\mathcal{N}=1/2$) at every interior θ. The two ports swap roles at $\theta\approx 52.4^\circ$, and the other axis becomes a product state exactly once, at cot $\theta=\eta$ ($\theta\approx 65.7^\circ$), while the plateau holds.
- So the success effect $E_s=K_{\rm coll}^\dagger K_{\rm coll}$ never factorizes: the optimal filter asks a genuinely **joint question**, and implementing it requires a two-qubit entangling operation; two CNOTs suffice (the $1\to 2$ isometry form of V, 11 parameters).
- The product class of panel (b) is exactly this plot's $\mathcal{N}=0$ class, and it is the additive one.



Negativity $\mathcal{N}$ of the optimized filter's accepted axes versus θ; the horizontal line is the Bell value $1/2$.

MESSAGE

What goes into the paper, and what waits

Single-sensor optimal-filter design extends to collective filter design for two sensors. The maximal enhancement is not uniquely tied to initial-state entanglement; within our numerical optimization, the clear distinction is between product and collective non-product filters.

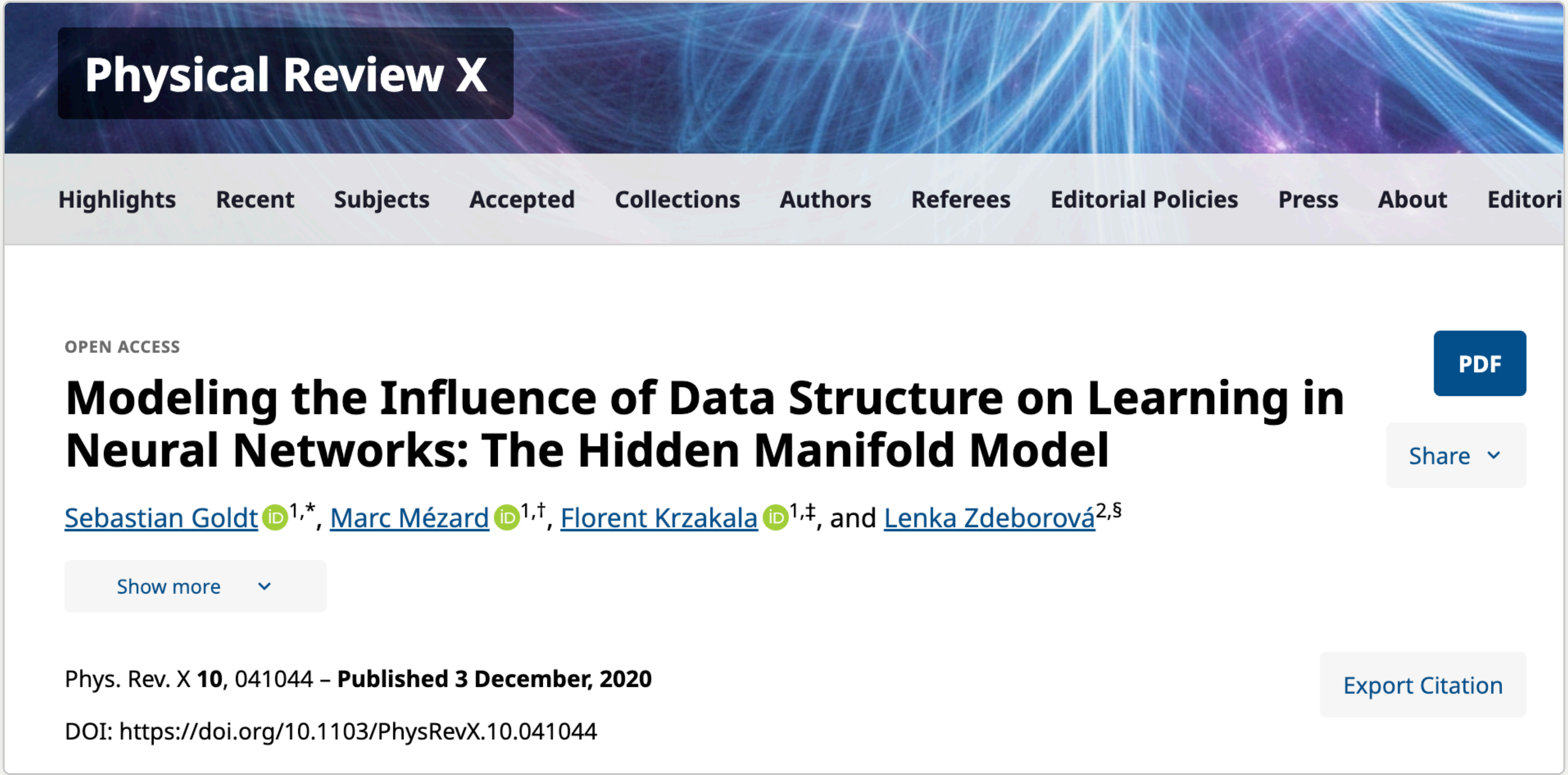
- **Reading.** The collective filter performs a *metrological information compression*: it concentrates the information of both sensors into the accepted branch ($4\times$ per accepted event), where independent filtering only ever collects the additive sum.
- **Deliberately postponed to a follow-up letter:** the analytic proof of the product-filter (additive) bound, the precise condition on V for collective enhancement, and the Kirkwood-Dirac / nonclassicality interpretation. The paper leads with the numerical $4\times$.
- **Status.** The two-sensor storyline is agreed, the composite figure (a to d) is implemented with the campaign data, and the section bullet points are drafted; what remains is turning them into section text.

02

The hidden manifold model

The June 29 journal-club material, revisited, and what has moved since

The paper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Modeling the Influence of Data Structure on Learning in Neural Networks: The Hidden Manifold Model

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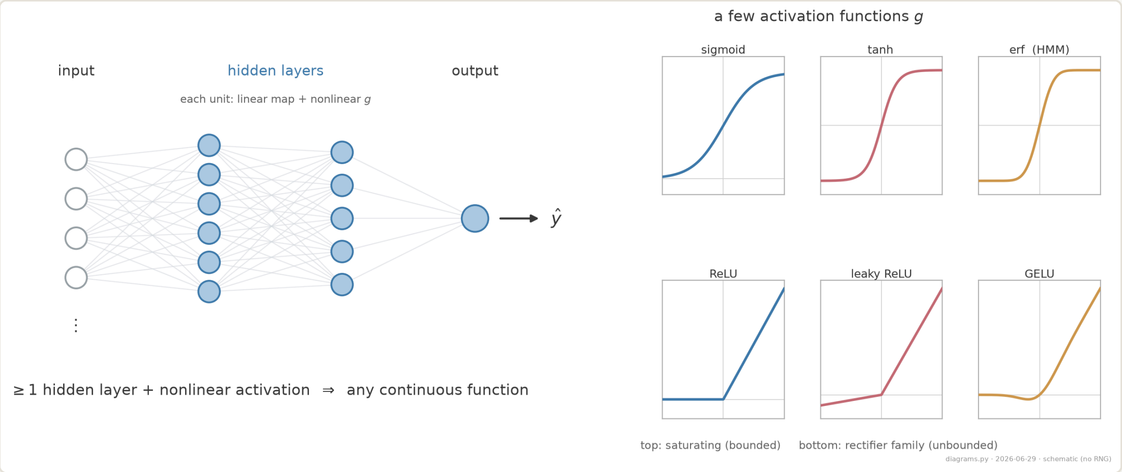
INTRO Neural networks won data science

- **Malleable and scalable.** Stacking and widening layers lets the *same* architecture fit vision, language, and control.
- **One universal building block.** A layer is $x \mapsto g(Wx)$: a linear map followed by a pointwise nonlinearity.
- **Mature tooling.** Autodiff frameworks (**PyTorch**, **TensorFlow**) made training networks the default in data science and modern AI.



Schematic feed-forward net (layers 4-6-5-1, illustrative): each layer is a linear map followed by a pointwise activation.

INTRO Why they work, in principle

Universal Approximation Theorem. A feed-forward network with

- at least one hidden layer,
- enough hidden units, and
- a nonlinear activation g (sigmoid, $\tanh$, ReLU),

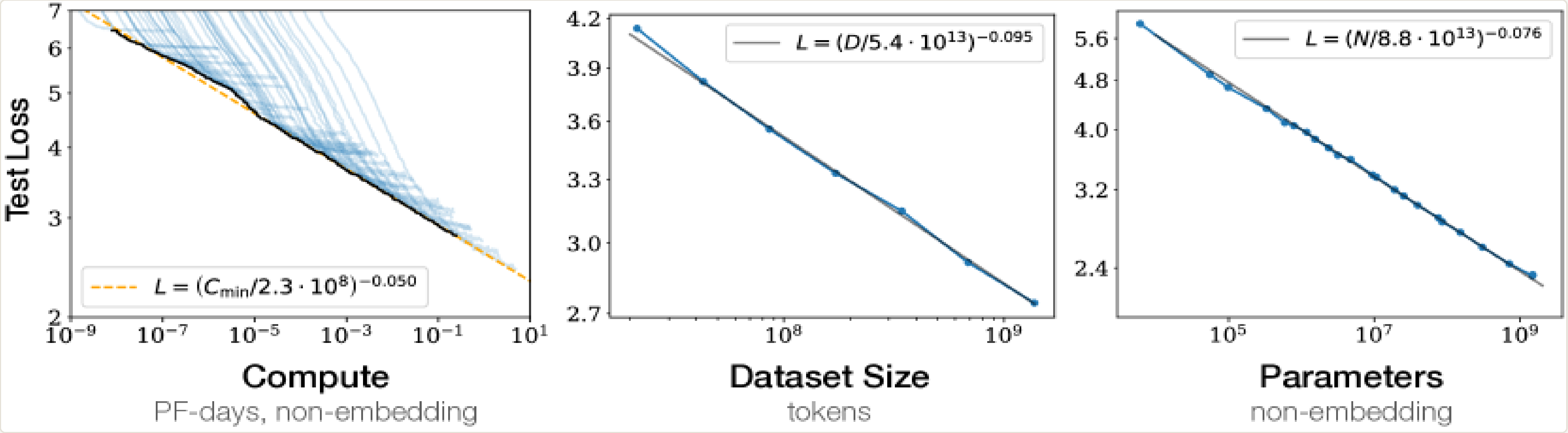
can approximate any continuous function on a compact subset of $\mathbb{R}^n$ to arbitrary accuracy.

Training also works in practice: given a well-posed dataset and a convex surrogate loss (squared error or cross-entropy), SGD reliably finds a good fit.

> **The open question.** Existence and trainability tell us *that* a network can fit, not *how* it learns or *why* it generalizes.

INTRO We do not understand the behavior

- For small models, trial-and-error was enough: tune width and hyperparameters by hand.
- As models became large and expensive, that approach stopped scaling, which motivated a "physics for AI": treat the network as a many-body system and look for laws.
- A leading example is **LLM scaling laws**: the test loss falls as a power law in compute, dataset size, and model size, a regularity we can measure but not yet derive. (Kaplan et al. 2020)



Kaplan et al. 2020, Fig. 1: test loss falls as a power law in compute, dataset size, and parameter count (WebText2 language models; their published figure).

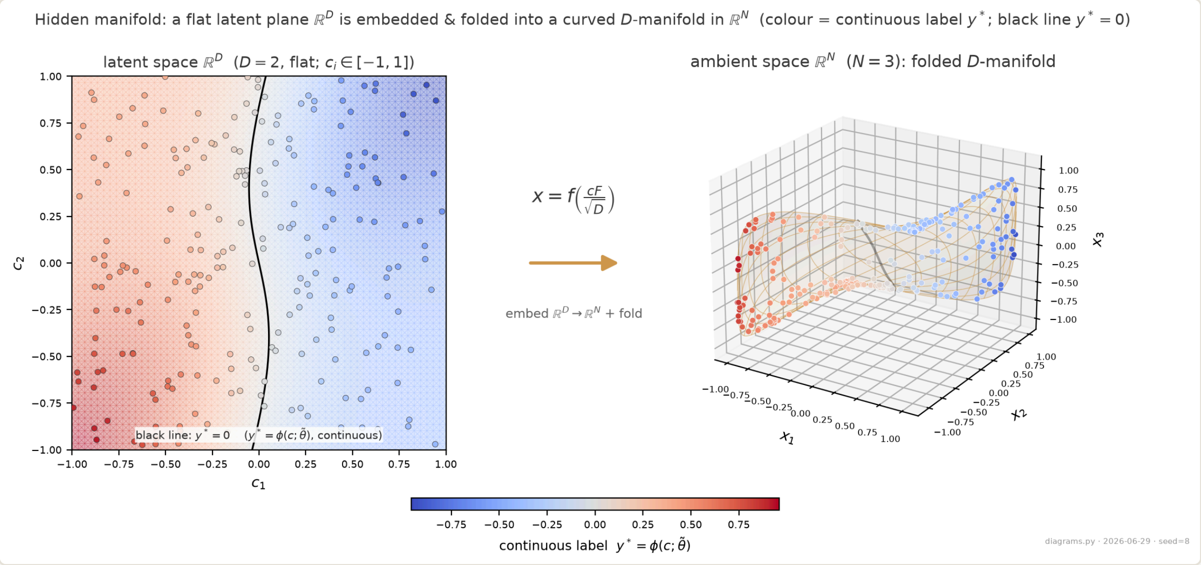
INTRO

This paper: statistical physics of the dynamics

- Statistical physics usually describes stationary states through a few **order parameters**, macroscopic summaries of a many-body system.
- This paper instead tracks order parameters **and their time evolution**: equations of motion for learning.
- Here "time" is training progress (the number of samples seen). The claim is that a few macroscopic overlaps evolve by **closed ODEs** that can be written down and integrated.

INTRO Why this is possible: solvable data

- Real data such as images is natural and, in ML benchmarks, deliberately decorrelated, which makes the learning dynamics analytically opaque.
- The hidden manifold model replaces this with data that has a controlled statistical structure (the **Gaussian Equivalence Property**, GEP), which makes the dynamics solvable.
- The data is synthetic, but it lets us follow exactly what happens during training, and in principle in more complex models.



A flat latent plane folded into a curved manifold; colour = continuous label y.*
Illustration: latent D=2 to ambient N=3, teacher M=3, inputs bounded to [-1,1], seed 8.

INTRO Why I care: quantum machine learning

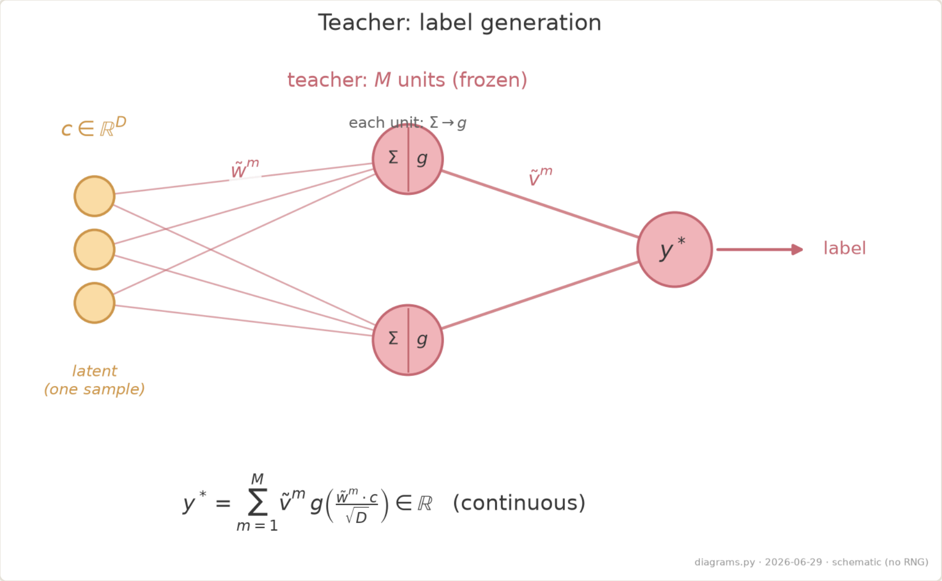
- Our **DQML** project has no native dataset and no large quantum hardware, so we emulate small circuits on classical GPUs.
- A frequent claim for QML is that it needs far fewer parameters than classical models.
- The benchmark study **"Better than classical?"** (Bowles, Ahmed & Schuld 2024) questions this claim:

> **Findings.** Out-of-the-box **classical models outperform all 12 quantum models** across 160 benchmark datasets, and **removing entanglement rarely changes the result**, so "quantumness" may not be the deciding factor. The tasks are easy binary classification, and the quantum models often reproduce classical ones.

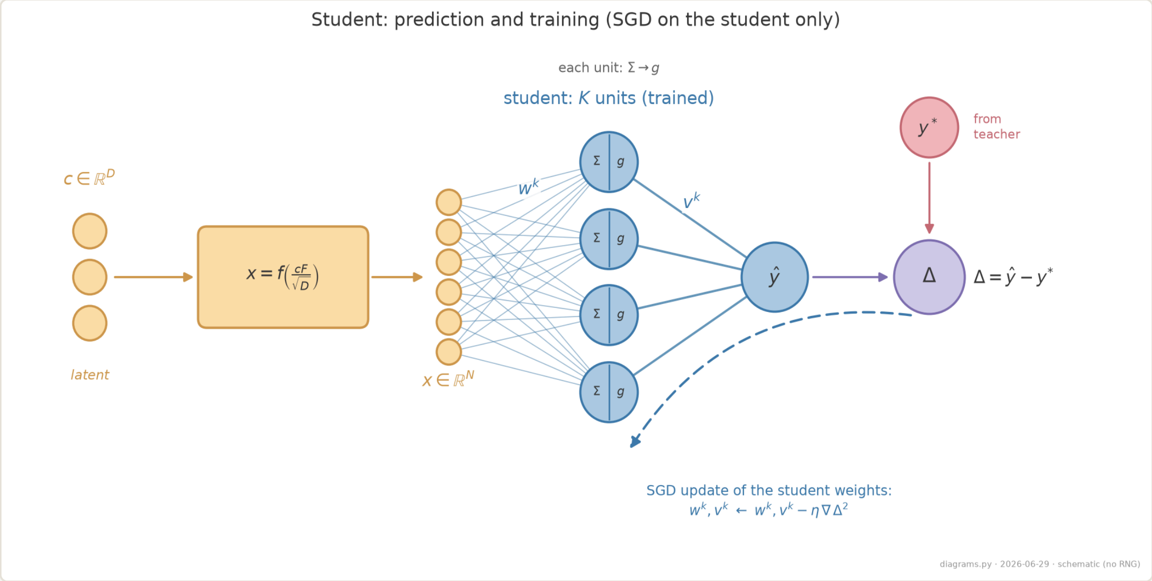
- We need data and analyses that identify where quantum methods actually help. An analytic handle of the HMM type is one such tool.

MODEL Structured data from a generator

- A generative model maps low-dimensional latent noise to high-dimensional, realistic-looking data, as a GAN generator does. The HMM uses the simplest such map, a single nonlinear transformation, with a **teacher** that assigns the labels.
- This is a **teacher-student** setup: the teacher fixes the ground-truth labels, and the student is trained to match them. The two need not be the same size.



Teacher (frozen): maps the latent c to the continuous label y. Schematic; M=2 units drawn.*



Student (trained): maps the folded input x to a prediction; the prediction error (prediction minus target) drives SGD on the student only. Schematic; K=4 units drawn.

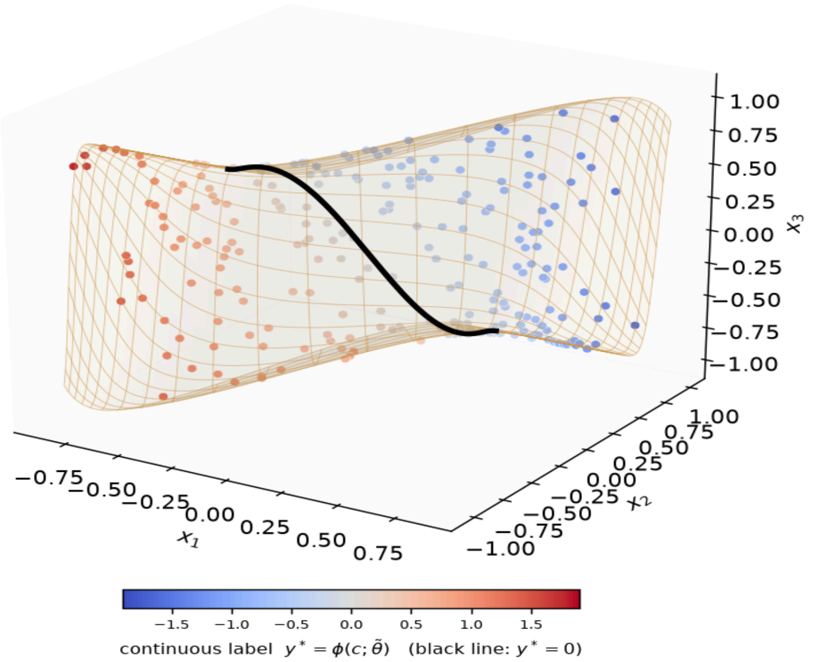
MODEL The hidden manifold

- Latent coordinates c of dimension D, drawn i.i.d. Gaussian.
- These are folded into a high-dimensional ambient space ($N \gg D$) through a fixed feature matrix F and a pointwise nonlinearity f:

$$x = f\left(\frac{cF}{\sqrt{D}}\right) \in \mathbb{R}^N.$$

- The labels depend only on c through the teacher. The student sees only x, so it must recover the manifold structure. In this sense the manifold is hidden.

Fig. 2 (HMM) — inputs $x = f(cF^\top/\sqrt{D})$ ($D = 2$, $N = 3$) concentrate on a lower-dimensional manifold; the label y^* is a function of position on it



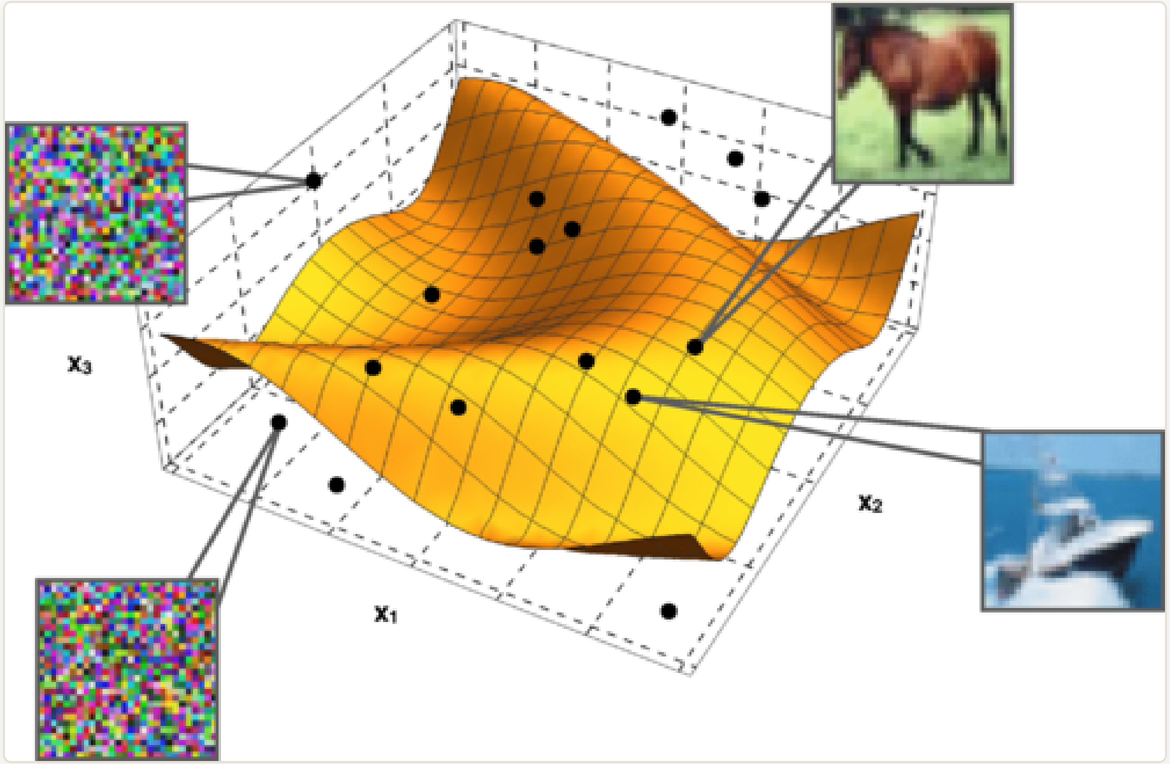
figures.py · 2026-06-29 · {'seed': 2, 'D': 2, 'N': 3, 'M': 2, 'fold': 'erf', 'label': 'continuous', 'note': 'illustration'}

Folded inputs concentrate on a 2D sheet in 3D space; colour = teacher label. Illustration: latent D=2, ambient N=3, teacher M=2, seed 2 (the real target is continuous).

MODEL

Why this is close to real data

- **Manifold hypothesis.** Realistic data (CIFAR-10, MNIST) does not fill its pixel space; it concentrates on a low-dimensional manifold.
- A few features, for example **PCA** components, already separate the classes.
- The HMM is a minimal instance of this picture: recognizable images lie on the surface, and off-manifold points are noise.



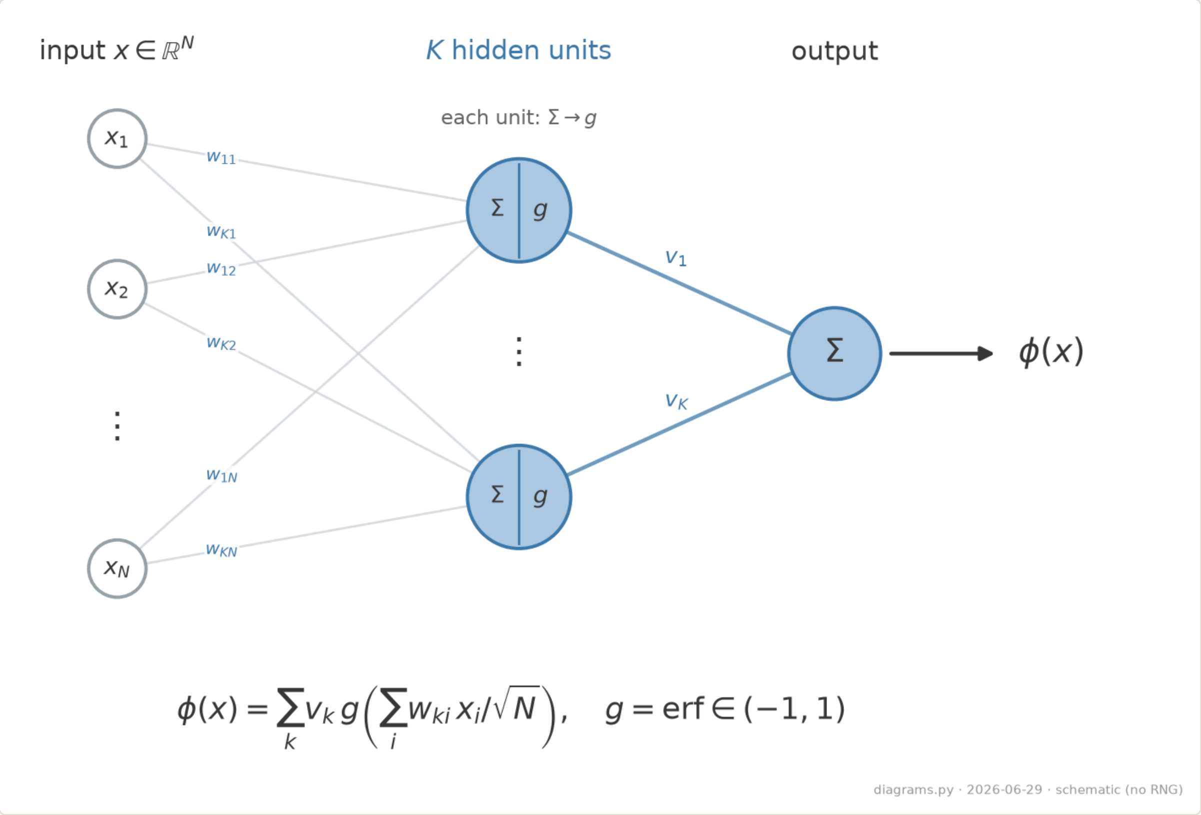
Goldt et al. Fig. 1 (conceptual illustration): recognizable CIFAR images lie on the manifold; off-surface points are random noise.

Two-layer student, online SGD

Student (K hidden units):

$$\phi(x) = \sum_{k=1}^{K} v_k \, g\left(\frac{w_k \cdot x}{\sqrt{N}}\right).$$

- A **teacher** (M units) on the latent c sets the target y^*; the student is trained by one-pass **online SGD** (rate η), with $\alpha = P/N$ and $\delta = D/N$ held fixed.
- Bounded hidden units ($g = \mathrm{erf} \in (-1, 1)$), but a **linear readout**, so the output is an *unbounded continuous real number*: the task is **regression** (MSE), not classification.



The two-layer (soft-committee) student: each unit applies the activation to a weighted sum, and a linear readout combines them. Schematic; K=2 units, N=3 inputs drawn.

SETUP The learning step (online SGD)

The student learns **one sample at a time**. At step μ:

1. Draw a **fresh** pair (x_μ, y_μ^*), with the label $y_\mu^* = \phi(c_\mu; \tilde{\theta})$ set by the teacher.
2. Predict $\hat{y} = \phi(x_\mu; \theta)$ and measure the error $\Delta_\mu = \hat{y} - y_\mu^*$.
3. Take one gradient step on the squared error $\frac{1}{2}\Delta_\mu^2$ (learning rate η):

$$\theta_{\mu+1} = \theta_\mu - \eta \nabla_\theta \tfrac{1}{2}\Delta_\mu^2, \qquad \theta = (W, v).$$

- **One pass:** each sample is used once, then discarded, so the weights stay independent of the next sample. (We write this step out weight-by-weight later, to turn it into the equations of motion.)
- We grade the student by the **test (generalization) error** on unseen data, $\epsilon_g = \frac{1}{2}\,\mathbb{E}[(\hat{y} - y^*)^2]$, not the training loss.
- This generalizes the classic **teacher-student** setup (i.i.d. Gaussian inputs); the HMM keeps that solvability but adds the manifold structure real data has.

THEORY

Local fields

A neuron's output depends on its input only through its **pre-activation**, the scalar that enters the activation function. These pre-activations are the **local fields**:

$$\lambda^k = \frac{1}{\sqrt{N}} \sum_i w_i^k \, f(u_i), \qquad \nu^m = \frac{1}{\sqrt{D}} \sum_r c_r \, \tilde{w}_r^m, \qquad u_i = \frac{1}{\sqrt{D}} \sum_r c_r F_{ir}.$$

- u_i: the pre-activation of input feature i, the latent vector projected onto row i of F, before folding by f.
- λ^k: the pre-activation of **student** hidden unit k, a weighted sum of the folded inputs $f(u_i)$.
- ν^m: the pre-activation of **teacher** hidden unit m, a weighted sum of the latent coordinates.

The factors $1/\sqrt{N}$ and $1/\sqrt{D}$ keep each field of order one. The test error depends on the data only through these $K + M$ scalar fields.

THEORY The Gaussian Equivalence Property

Property (GEP). In the limit $N, P, D \to \infty$ (with α, δ fixed), the $K + M$ local fields $\{\lambda^k\}, \{\nu^m\}$ are **jointly Gaussian**, so their joint distribution is fixed entirely by their first two moments.

The means are $\mathcal{O}(1/\sqrt{N})$ for the student fields and 0 for the teacher fields; the covariances are the three overlap matrices on the next slide.

> **Consequence.** Everything the test error can depend on is contained in a few **covariances**, the order parameters; the individual microscopic weights drop out.

THEORY

Order parameters: the overlaps

Each order parameter is a covariance of the (centred) local fields:

$$Q^{k\ell} = \mathbb{E}[\bar{\lambda}^k \bar{\lambda}^\ell], \qquad R^{km} = \mathbb{E}[\bar{\lambda}^k \nu^m], \qquad T^{mn} = \mathbb{E}[\nu^m \nu^n].$$

- $Q^{k\ell}$ (**student-student**): how aligned student units k and ℓ are. The diagonal Q^{kk} is the variance, i.e. the size, of unit k's field.
- R^{km} (**student-teacher**): how much student unit k has aligned with teacher unit m. Learning is the growth of these overlaps; **specialization** means each k locks onto a single m.
- T^{mn} (**teacher-teacher**): fixed by the frozen teacher, it encodes the structure of the target.

The generalization error is a function of Q, R, T and the second-layer weights v alone.

THEORY

Order parameters: the reduction

The student-side overlaps come from two underlying matrices:

$$W^{k\ell} = \frac{1}{N}\sum_i w_i^k w_i^\ell \ \text{(ambient weight overlap)}, \qquad \Sigma^{k\ell} = \frac{1}{D}\sum_r S_r^k S_r^\ell, \quad S_r^k = \frac{1}{\sqrt{N}}\sum_i w_i^k F_{ir}.$$

The folding function f enters **only** through three scalars (for $u \sim \mathcal{N}(0,1)$):

$$a = \langle f(u)\rangle, \qquad b = \langle u\, f(u)\rangle, \qquad c = \langle f(u)^2\rangle.$$

These combine into the student-side overlaps:

$$Q^{k\ell} = (c - a^2 - b^2)\, W^{k\ell} + b^2\, \Sigma^{k\ell}, \qquad R^{km} = b\,\frac{1}{D}\sum_r S_r^k\, \tilde{w}_r^m.$$

Any two folding functions with the same (a, b, c) produce identical learning curves. (For an odd folding $a = \langle f\rangle = 0$, so the dynamics below use $Q = (c - b^2)W + b^2\Sigma$.)

THEORY

From weight updates to equations of motion

Every equation of motion descends from the **SGD updates** of the weights (one fresh sample per step):

$$(w_i^k)_{\mu+1} - (w_i^k)_\mu = -\frac{\eta}{\sqrt{N}}\, v^k\, \Delta\, g'(\lambda^k)\, f(u_i), \qquad v^k_{\mu+1} - v^k_\mu = -\frac{\eta}{N}\, \Delta\, g(\lambda^k),$$

with the prediction error $\Delta = \sum_j v^j g(\lambda^j) - \sum_m \tilde{v}^m \tilde{g}(\nu^m)$.

The test error sees the data **only through the local fields**, so by the GEP it collapses to a function of the order parameters alone:

$$\epsilon_g = \tfrac{1}{2}\,\mathbb{E}\big[(\phi(x;\theta) - y^*)^2\big] = \tfrac{1}{2}\,\mathbb{E}\Big[\big(\sum_k v^k g(\lambda^k) - \sum_m \tilde{v}^m \tilde{g}(\nu^m)\big)^2\Big] \xrightarrow{N,D\to\infty} \epsilon_g\big(Q, R, T, v, \tilde{v}\big).$$

So we recast each weight update as the induced change in Q, R, T. Because a fresh sample is independent of the current weights, the per-step change **self-averages**; in the limit it becomes a deterministic ODE in the time $t = \mu/N$. (We set $a = \langle f \rangle = 0$, so $Q = (c - b^2)W + b^2\Sigma$.)

THEORY Equations of motion (1): direct overlaps

The dynamics are driven by Gaussian averages over the jointly-Gaussian fields (indices j, k, ℓ run over student units, m, n over teacher units; g_a is the activation of the third unit):

$$I_2(k,j) = \mathbb{E}[g(\lambda^k)g(\lambda^j)], \quad I_3(k,j,a) = \mathbb{E}[g'(\lambda^k)\,\lambda^j\,g_a], \quad I_4(k,\ell,j,\iota) = \mathbb{E}[g'(\lambda^k)g'(\lambda^\ell)g(\lambda^j)g(\lambda^\iota)].$$

Second-layer weights v^k:

$$\frac{dv^k}{dt} = \eta\Big[\sum_n \tilde{v}_n\, I_2(k,n) - \sum_j v^j\, I_2(k,j)\Big].$$

Ambient student-student overlap $W^{k\ell} = \frac{1}{N}\sum_i w_i^k w_i^\ell$:

$$\frac{dW^{k\ell}}{dt} = -\eta v^k\Big(\sum_j v^j I_3(k,\ell,j) - \sum_n \tilde{v}^n I_3(k,\ell,n)\Big) - \eta v^\ell\Big(\sum_j v^j I_3(\ell,k,j) - \sum_n \tilde{v}^n I_3(\ell,k,n)\Big)$$
$$+ c\,\eta^2 v^k v^\ell\Big(\sum_{j,\iota} v^j v^\iota I_4(k,\ell,j,\iota) - 2\sum_{j,m} v^j \tilde{v}^m I_4(k,\ell,j,m) + \sum_{n,m} \tilde{v}^n \tilde{v}^m I_4(k,\ell,n,m)\Big).$$

THEORY

Equations of motion (2): spectral densities

The remaining overlaps become densities over the spectrum of $\Omega = \frac{1}{N} F F^\top$ (eigenvalues ρ), with $d(\rho) = (c - b^2)\delta + b^2 \rho$ and $R^{km} = b \int d\rho \, p_\Omega(\rho) \, r^{km}(\rho, t)$:

$$\frac{\partial r^{km}}{\partial t} = -\frac{\eta}{\delta} v^k \Bigg[d(\rho) \, r^{km} \sum_{j \neq k} v^j \frac{Q^{jj} I_3(k,k,j) - Q^{kj} I_3(k,j,j)}{Q^{jj} Q^{kk} - (Q^{kj})^2} + d(\rho) \sum_{j \neq k} v^j r^{jm} \frac{Q^{kk} I_3(k,j,j) - Q^{kj} I_3(k,k,j)}{Q^{jj} Q^{kk} - (Q^{kj})^2}$$

$$+ \frac{v^k \, d(\rho) \, r^{km}}{Q^{kk}} I_3(k,k,k) - d(\rho) \, r^{km} \sum_n \tilde{v}^n \frac{T^{nn} I_3(k,k,n) - R^{kn} I_3(k,n,n)}{Q^{kk} T^{nn} - (R^{kn})^2} - b\rho \sum_n \tilde{v}^n \tilde{T}^{nm} \frac{Q^{kk} I_3(k,n,n) - R^{kn} I_3(k,k,n)}{Q^{kk} T^{nn} - (R^{kn})^2} \Bigg].$$

The latent overlap $\Sigma^{k\ell} = \int d\rho \, p_\Omega(\rho) \, \sigma^{k\ell}(\rho, t)$ obeys a structurally identical equation. Together with $Q^{k\ell} = (c - b^2) W^{k\ell} + b^2 \Sigma^{k\ell}$ and the v^k, $W^{k\ell}$ equations above, the system is closed (no free parameters beyond a, b, c, η, δ).

THEORY

Closing the equations, and the test error

Three steps make the system finite and explicit:

1. **Eigenbasis.** Projecting onto the eigenvectors of $\Omega = \frac{1}{N} F F^\top$ replaces an infinite hierarchy of order parameters with densities over the eigenvalues ρ (Marchenko-Pastur for Gaussian F).
2. **Densities.** The overlaps become $r^{km}(\rho, t)$ and $\sigma^{k\ell}(\rho, t)$, integrated against the spectrum.
3. **Gaussian integrals.** I_2, I_3, I_4 have closed forms (arcsin, arctan) for erf activations.

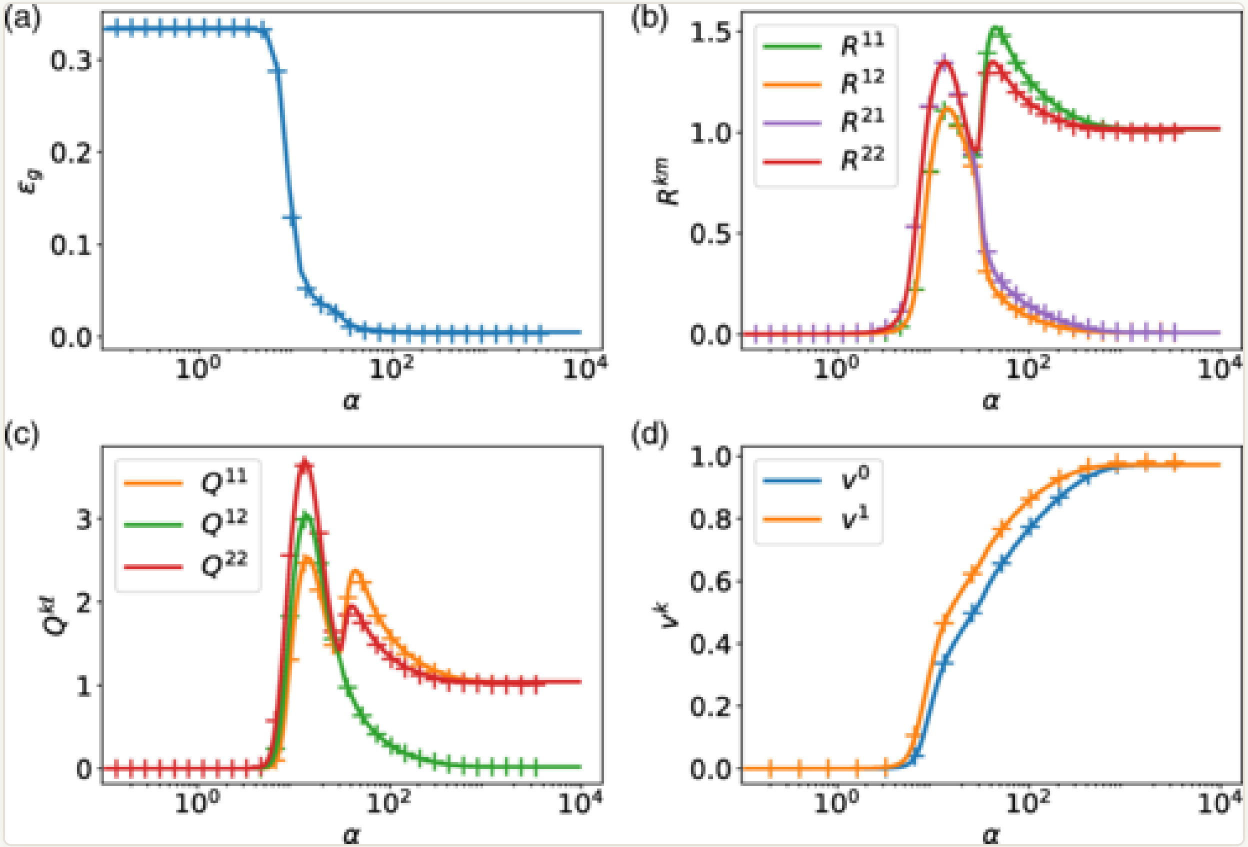
Integrating the system and substituting back gives the closed-form generalization error:

$$\epsilon_g = \frac{1}{\pi} \sum_{k,\ell} v^k v^\ell \arcsin \frac{Q^{k\ell}}{\sqrt{1+Q^{kk}}\sqrt{1+Q^{\ell\ell}}} - \sum_{k,n} v^k \tilde{v}^n \frac{R^{kn}}{\sqrt{2\pi}\sqrt{1+Q^{kk}}} + \text{(teacher term)}.$$

RESULT

The learning curve

- Online dynamics versus $\alpha = P/N$ (which plays the role of training time): the order parameters and ϵ_g trace three phases.
- A plateau, then **specialization** (each student unit locks onto a teacher unit), then an asymptote.
- The **ODE prediction** (lines) sits on top of the finite-size **simulation** (crosses), with no fitting.

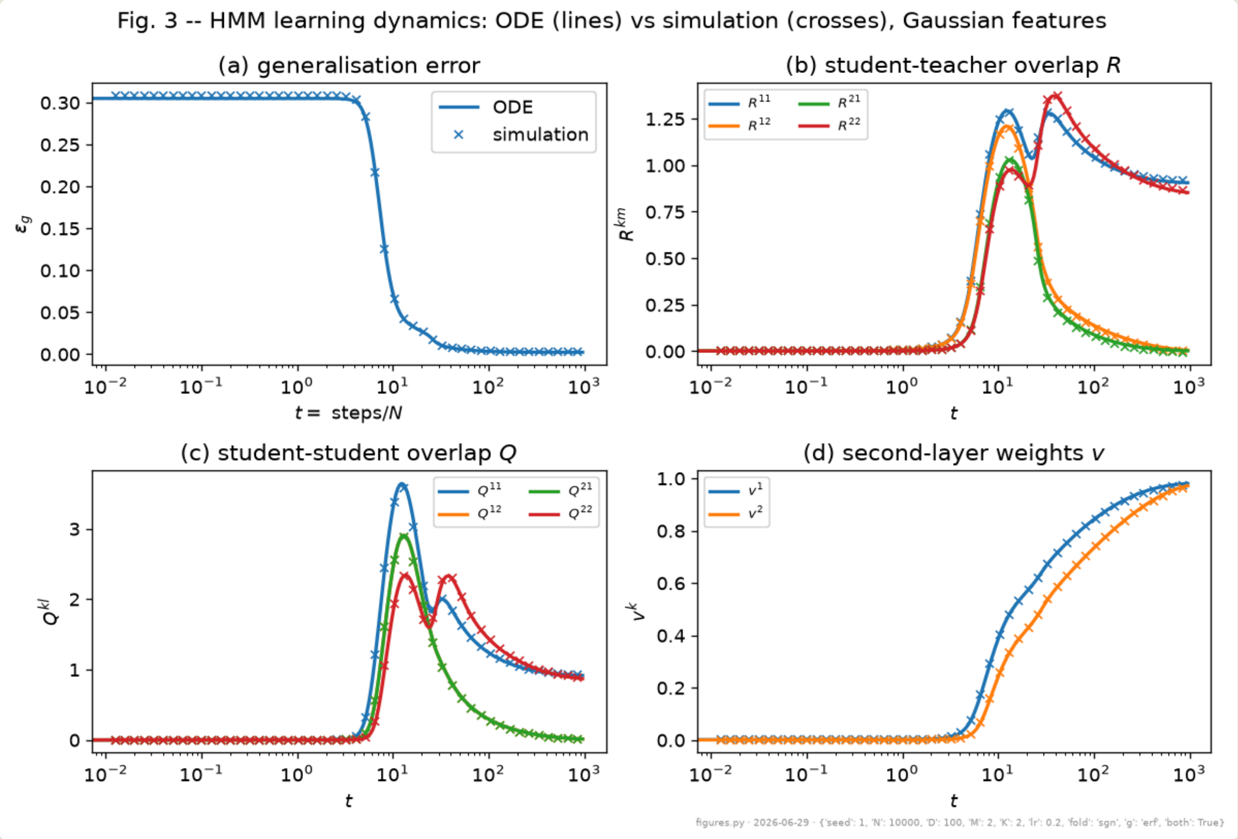


Goldt et al. Fig. 3: test error and order parameters vs. time t=steps/N; theory (lines) vs. simulation (crosses). N=10,000, D=100, K=M=2, rate 0.2, sign folding / erf activation.

RESULT

Reproduction, from scratch

- A from-scratch **PyTorch** port of both the simulator and the ODE integrator.
- It tracks a single simulation to $\max|\Delta\epsilon_g| \approx 9 \times 10^{-4}$ across the whole trajectory, reproducing the paper's central claim.
- The small residual is only the difference between Python and C++ random number generation, not a modeling gap; it is cross-checked against the authors' C++.

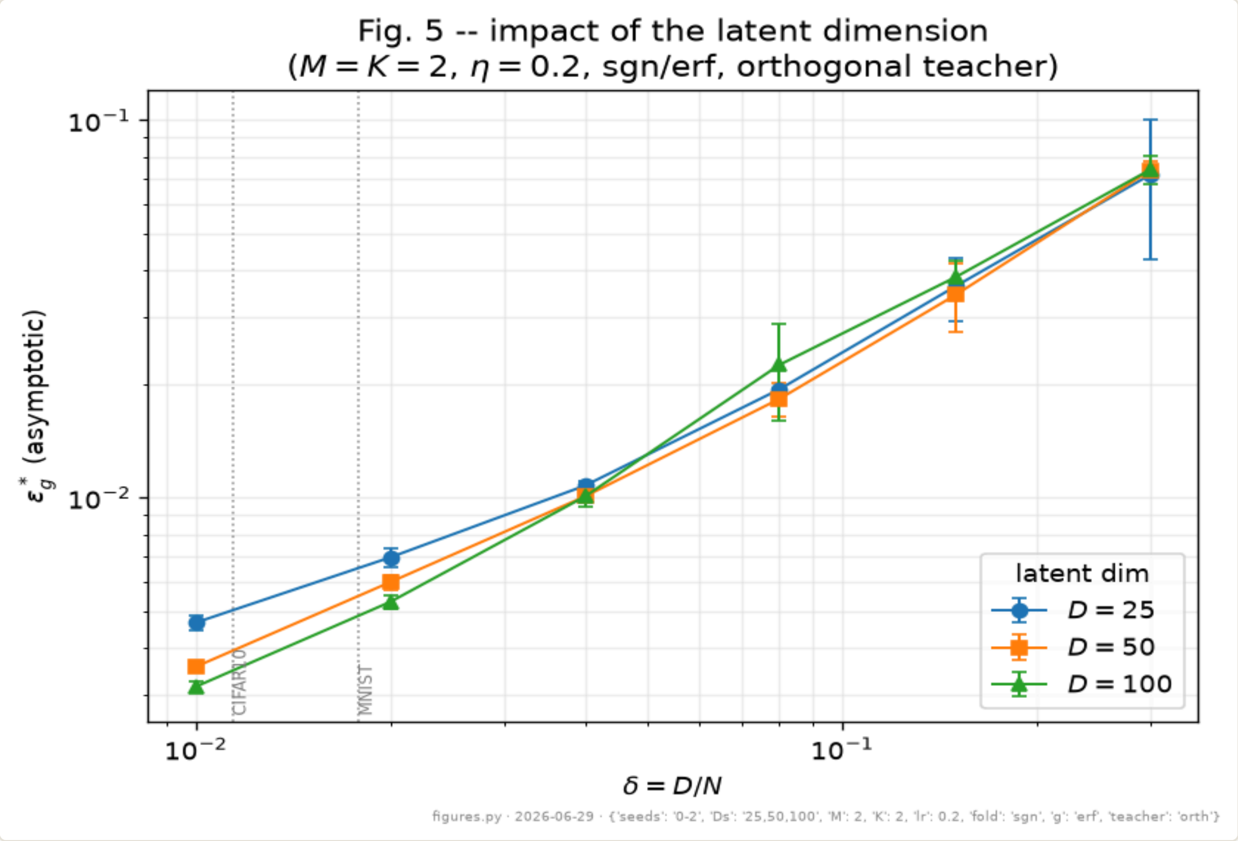


Our PyTorch reproduction at the same settings: ODE (lines) vs. one simulation (crosses), agreeing to about 1e-3. N=10,000, D=100, K=M=2, rate 0.2, sign/erf, seed 1.

RESULT

Structure helps generalization

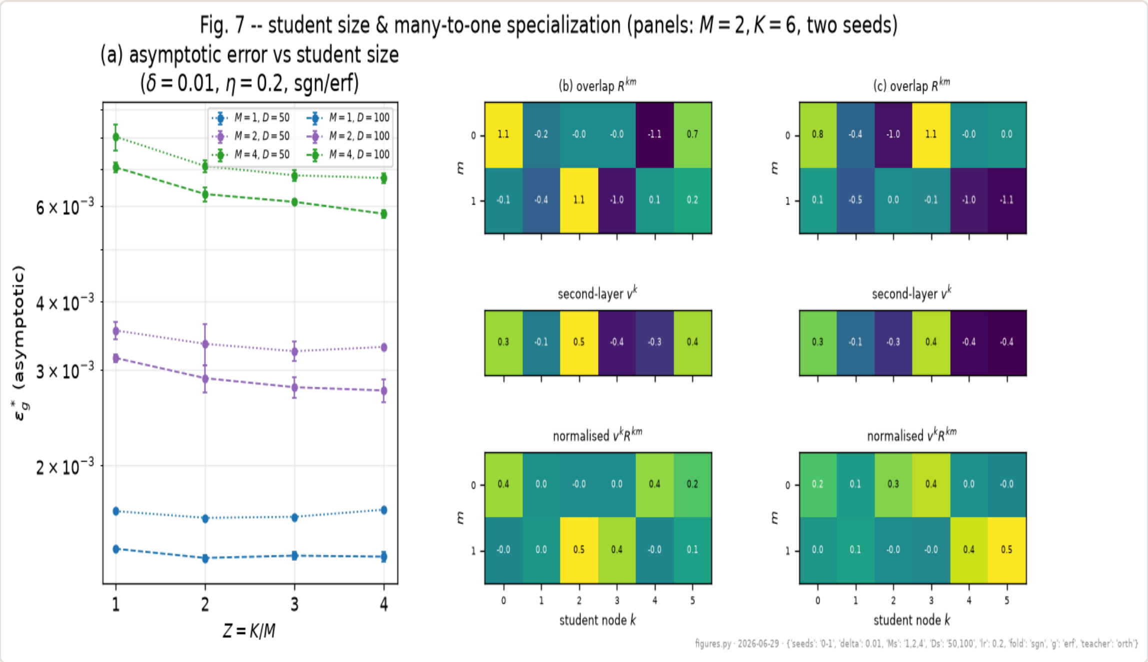
- Asymptotic error ϵ_g^* versus $\delta = D/N$, the intrinsic-dimension ratio (log-log axes).
- A lower intrinsic dimension gives better generalization: a more sharply folded, lower-dimensional manifold is easier to learn.
- Real datasets sit at small δ; the markers indicate MNIST and CIFAR for context.



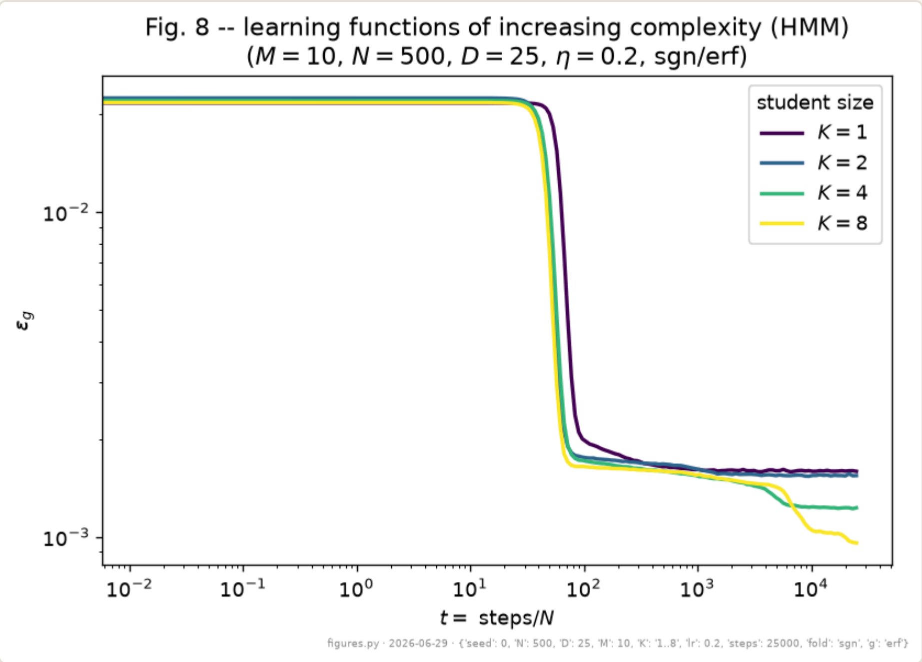
Asymptotic test error vs. the dimension ratio D/N: a smaller latent dimension generalizes better. K=M=2, rate 0.2, sign/erf, orthogonalized teacher; D=25/50/100, mean over 3 seeds.

Specialization and model size

Larger students specialize many-to-one, and as complexity grows they separate to lower error.



Asymptotic error vs. student size K/M (teachers M=1,2,4; D=50,100; D/N=0.01, rate 0.2, sign/erf): student units specialize many-to-one, and gains saturate.



Learning curves for growing student size K=1..8 (teacher M=10, N=500, D=25, rate 0.2, sign/erf): wider students break away later, fitting functions of increasing complexity.

DISCUSSION

Critique

Elegant and exact, but obtained under strong simplifications:

- **A deliberately simple model.** A single hidden layer and plain SGD, chosen so the equations of motion close.
- **Online, not batch.** One-pass SGD with a fresh sample per step is atypical of modern (multi-epoch, mini-batch) training. Reusing samples might break the GEP.
- **A non-general task.** The target is a continuous, unbounded real number (MSE regression on a sum-of-sigmoids teacher with a linear, unsquashed readout), not the bounded or discrete classification that dominates real ML. (The two-colour manifold picture is for illustration.)
- **Not always a reduction.** The order parameters grow as $Q\,(K \times K)$, $R\,(K \times M)$, $T\,(M \times M)$, plus the spectral densities $r(\rho), \sigma(\rho)$. For a large student that is as many quantities as the model has parameters, so the "few macroscopic variables" framing is misleading.

OUTLOOK Toward a quantum version

For the DQML project, the HMM is a starting point:

- We would need a complex-valued version of the dataset, with structure native to quantum states and circuits rather than real images.
- We would need to derive our own equations of motion for the quantum model. Our current setup (a CNN with classical communication) is heavy to simulate and not sure to have closed dynamics.
- The payoff is an analytical tool to argue where quantumness helps, which addresses the question that **"Better than classical?"** poses.

> To demonstrate a real quantum advantage we need both data and theory that isolate the quantum resource. The HMM shows what solvable structured data can look like.

Summary

- The **hidden manifold model** generates structured data: a low-dimensional latent variable, folded into high dimensions, labeled by a teacher.
- The **Gaussian Equivalence Property** reduces two-layer online-SGD learning to **closed ODEs** for a few order parameters, giving the exact learning curve.
- Results: smaller δ (more structure) helps; units specialize; wider students reach higher complexity; the curve is reproduced to $\approx 10^{-3}$.
- For us: a concrete example of "physics for AI", and a target to adapt to quantum machine learning.

UPDATE Since the journal club: the HMM enters DQML

The outlook slide asked for a quantum-native version of this program. Two of its three levels now exist (July 15):

- **Level 0, the dataset bridge (built).** The HMM generator now emits DQML's exact train/validation/test format, with labels binarized at the teacher median for the cross-entropy classifier. The intrinsic-dimension ratio $\delta = D/N$ becomes the data-structure knob, and the correlation the HMM injects across QPUs is exactly its input covariance, $\mathrm{Cov}(x_i, x_j) = b^2(FF^\top/D)_{ij}$, so inter-QPU correlation is now analytic and designable.
- **Level 1, frozen circuit plus trained head (derived and verified).** Training only the readout head on a frozen circuit is a kernel / GLM problem on the quantum feature map: each Pauli feature decomposes exactly as $g_P(z) = 2^{-n} \sum_b \chi_P(b)\, e^{i\Delta_b(z)}$ with Δ_b quadratic in the inputs, so the feature means, the kernel, and the teacher alignment all have closed forms through one Gaussian integral. Verified against Monte Carlo; the feature identity holds to 6×10^{-16}.
- **Level 2, the trained deep circuit, stays open.** That is where the equations-of-motion question of the paper returns for the quantum model.

The practical consequence: the parameter sweeps can now target regimes the closed forms point at, instead of sweeping blind.

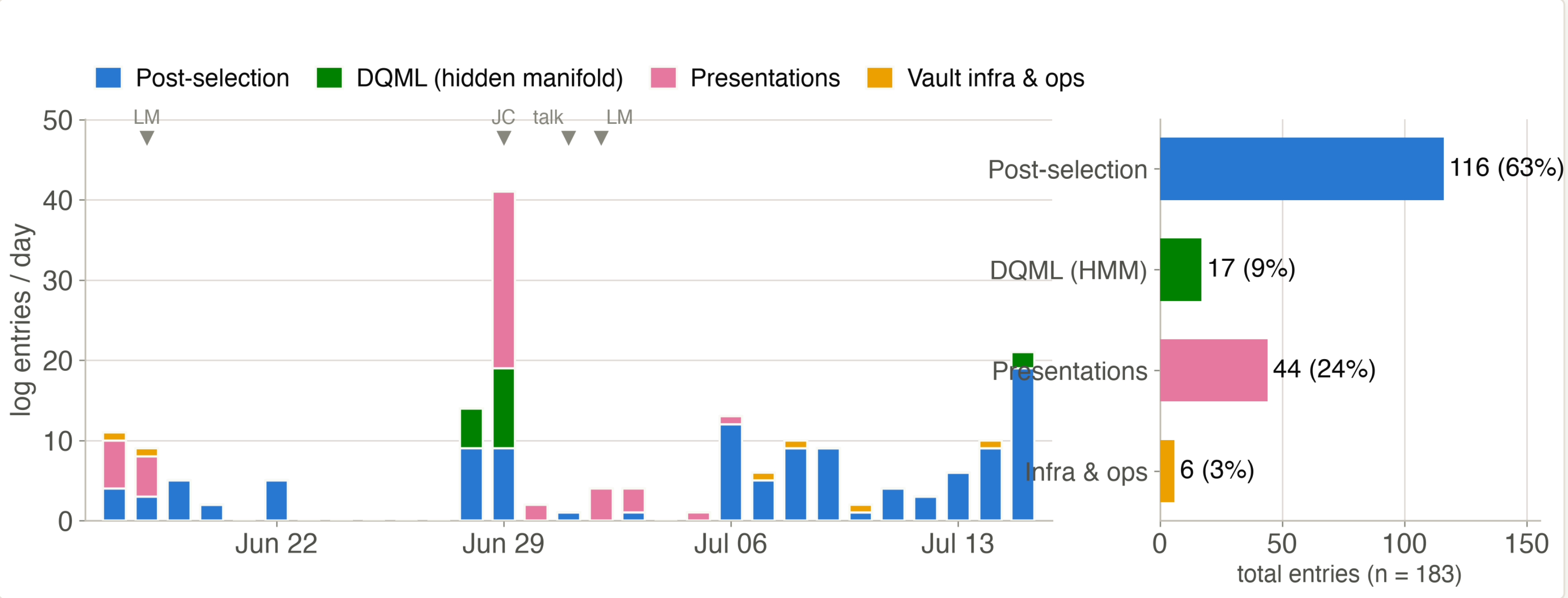
03

What I have done

Four weeks of logged work, against the June plan

REVIEW

Where the time went



Research-log entries per day, June 17 to July 15 (183 entries on 22 active days), classified by project. Markers: LM = lab meeting, JC = journal club, talk = the July 1 self-introduction. Entry counts are a proxy for time, not hours.

- **Post-selection took 63 %** and ran almost every active day; **presentations (24 %)** cluster right before each talk date.
- **DQML (9 %)** came in two bursts, late June and July 15; **DDrf never appears** (advisory work leaves no vault trace).

REVIEW Against the June plan

done **Post-selection, two-sensor.** Planned: wrap the paper content, bullet points first, figure within the week. Delivered: the storyline is agreed, the campaign is certified, and the composite figure (a to d) is implemented.

overshot **Post-selection, multi-qubit letter.** Planned: cross-validate the two filter derivations and outline a short letter by end of June. The filters turned out to be *literally the same operator*, and the analysis went far beyond the plan (N^2 certified at $N = 3, 4, 5$, closed forms, a project zip-up), but the letter itself is **not drafted**: much of July went into follow-up-paper material instead of writing.

wip **DQML.** Planned: understand the HMM and start simulations by June. Delivered: the reproduction, the journal club, and now the Level-0/Level-1 bridge. Still missing: the large parameter sweeps the plan called for.

todo **DDrf.** Advisory only, as planned (figure feedback, experiment discussions); the collaboration mode with Prof. Seo's group is still to be scheduled.

NEXT The push

The results for the paper exist; the bottleneck is now writing, so writing comes first.

- **Post-selection, this week.** Turn the agreed bullet points and the composite figure into the two-sensor section text of the draft. The supplement panels (optimal strengths, equal-strength classes, acceptance probability) are already built.
- **Post-selection, next.** Fix the follow-up letter to one or two figures: the N^2 scaling and the additive product bound. Most ingredients already exist in the vault, so this is scoping and writing, not new analysis.
- **DQML.** Move from derivation to simulation: sweep the data-structure knob $\delta = D/N$ with the Level-1 closed forms as targets, on the remote-server / Colab setup that is already in place.
- **DDrf.** Hold one planning discussion with Prof. Seo's group to fix the collaboration mode, since their interest leans toward DD rather than DDrf.

References

Post-selection

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2. S. Das, T. Modak, M. N. Bera, **"Saturating quantum advantages in postselected metrology with the positive Kirkwood-Dirac distribution,"** *Phys. Rev. A* **107**, 042413 (2023).
3. R. Iten, R. Colbeck, I. Kukuljan, J. Home, M. Christandl, **"Quantum circuits for isometries,"** *Phys. Rev. A* **93**, 032318 (2016).

Hidden manifold model

4. S. Goldt, M. Mézard, F. Krzakala, L. Zdeborová, **"Modeling the influence of data structure on learning in neural networks: The hidden manifold model,"** *Phys. Rev. X* **10**, 041044 (2020). arXiv:1909.11500.
5. J. Bowles, S. Ahmed, M. Schuld, **"Better than classical? The subtle art of benchmarking quantum machine learning models,"** arXiv:2403.07059 (2024).
6. G. Cybenko, *Math. Control Signals Systems* (1989); K. Hornik, *Neural Networks* (1991): the universal approximation theorems.
7. J. Kaplan *et al.*, **"Scaling laws for neural language models,"** arXiv:2001.08361 (2020).

Thank you

Questions & discussion

Donghun Jung

Research Update · Lab Meeting · 2026-07-16

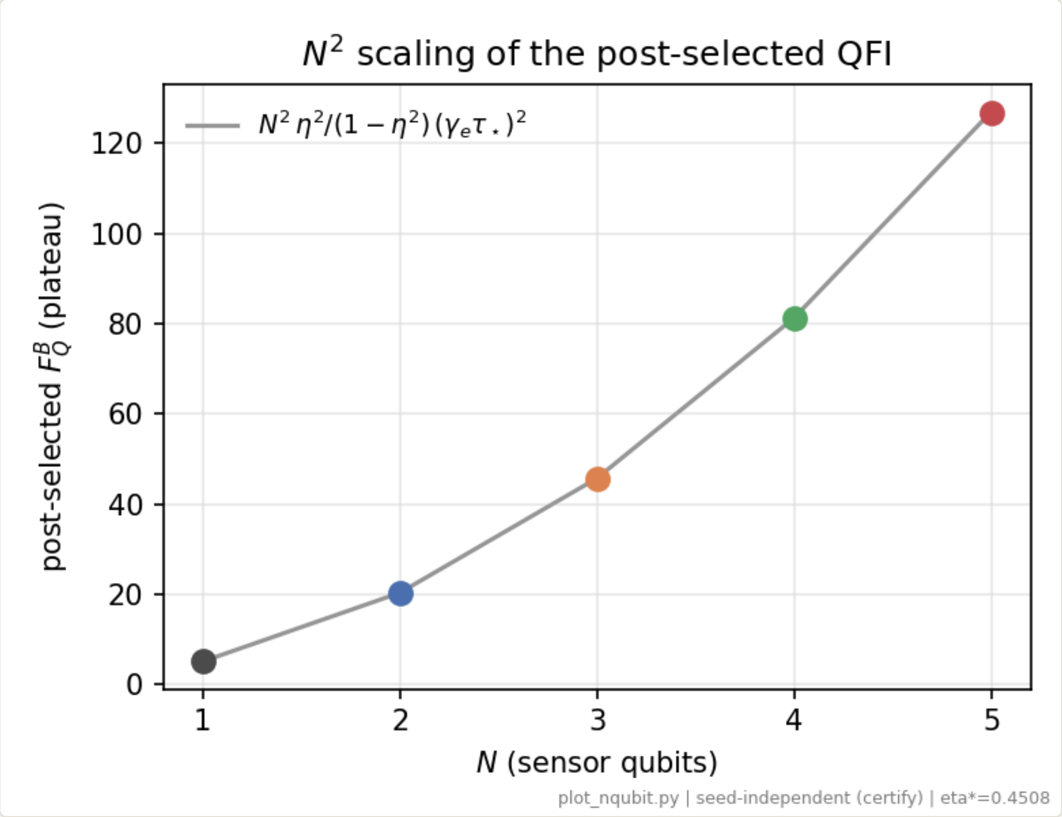
A

Backup

Extra material for questions

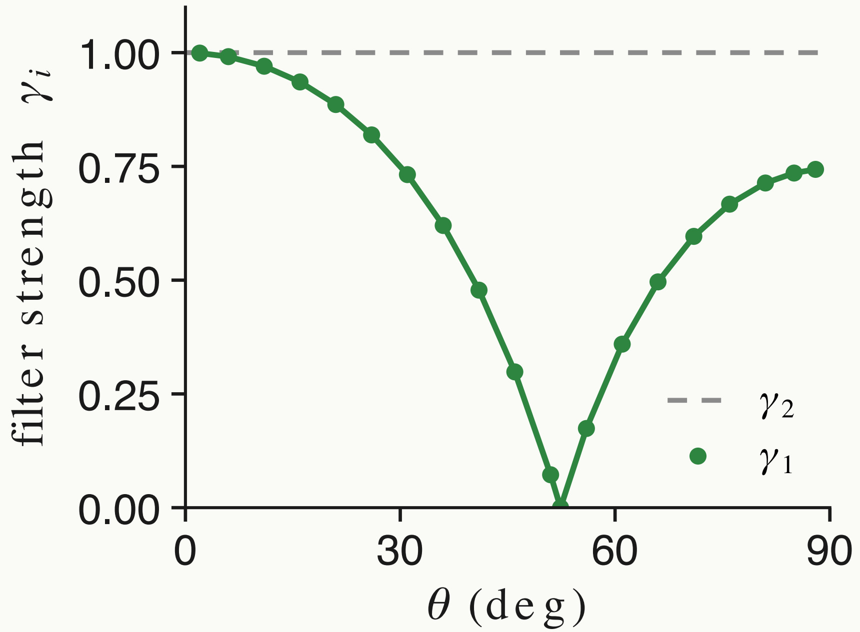
BACKUP Beyond two sensors: the N^2 scaling

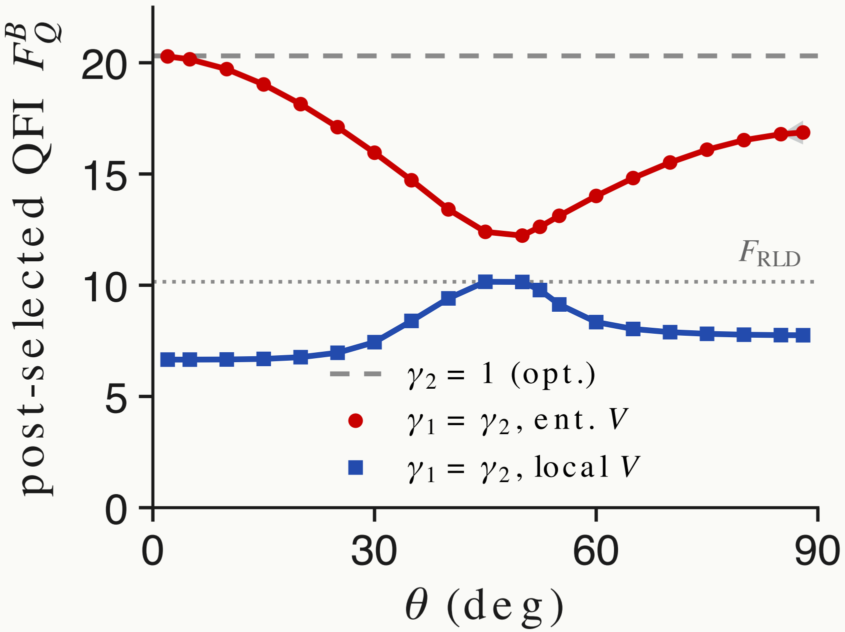
- The collective construction generalizes: at $N = 3, 4, 5$ sensors a continuation search certifies the plateau $F_Q^B = 45.70,\ 81.24,\ 126.90$, that is $9\times,\ 16\times,\ 25\times$ the single-sensor optimum.
- This matches the bound $F_Q^B = N^2\eta^2/(1-\eta^2)\,(\gamma_e\tau_\star)^2$: Heisenberg scaling in the sensor number, per accepted event, surviving dephasing.
- A free global search fails into a wrong basin from $N = 3$ on, which is why the certification needs warm starts; this scaling result is the core of the planned follow-up letter.



Certified plateau versus sensor number N; the line is $N^2\eta^2/(1-\eta^2)\,(\gamma_e\tau_\star)^2$.

BACKUP The optimal strengths, and what constraints cost



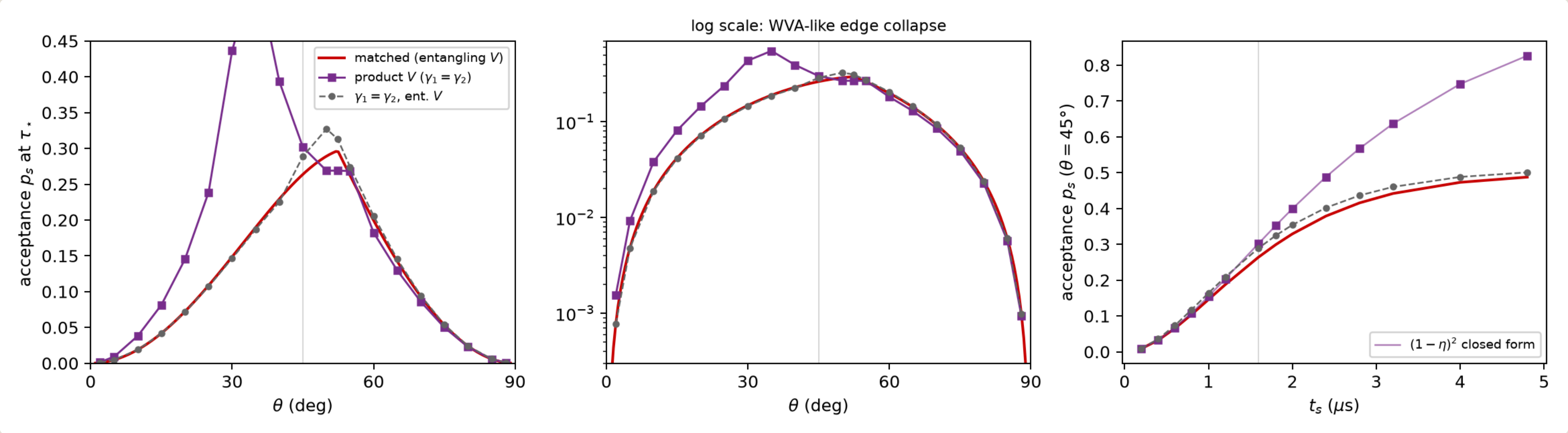


Matched filter: $\gamma_2 = 1$ always (one projective meter), while $\gamma_1 = 2\chi/(1+\chi)$ tracks the probe and touches 0 at $\theta \approx 52.4^\circ$.

Forcing equal strengths $\gamma_1 = \gamma_2$: with an entangling V (red) the QFI dips to about $60\,\%$ of the plateau near $\theta = 50^\circ$; with a local V (blue) it caps at the additive rung $F_{\mathrm{RLD}} = 10.15$.

The asymmetry $\gamma_2 = 1 \neq \gamma_1$ is not cosmetic: one meter interrogates sharply while the other is tuned to the probe, and constraining them equal costs real QFI.

BACKUP The price: acceptance probability



Acceptance probability p_s for the three filter classes: versus θ at $\tau_\star$ (left, and log scale, middle) and versus t_s at $\theta = 45^\circ$ (right).

- The matched (collective) filter accepts at most about 0.30 of runs, near $\theta \approx 52^\circ$, and its acceptance collapses toward the edges; the ceiling is $p_s \le (1-\eta^2)/N$ when the QFI plateau is attained.
- The product filter buys its smaller QFI with much higher acceptance at small θ (a weak-value-amplification-like trade, log panel); the paper states the per-accepted-event result and prices the yield separately.